Push and Turn

A Two-Channel Account of How Structure Forms, from a Collision Regime to a Computation Regime

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Abstract

We describe the formation of ordered structure as the interaction of two channels that run at the same time on the same trajectory, not as two eras that follow one another. The first channel is *push*: external forcing that drives a system, deposits energy, and builds or reshapes the constraints the system must satisfy. The second is *turn*: the internal evaluation a system performs against its own constraints, executed as a coordinate rotation that preserves an invariant while changing its representation. Push pays an energy tax and cannot be avoided if anything is to happen at all; turn cannot begin without a substrate that push has already laid down. Neither replaces the other. Along a single control parameter — the ratio of forcing bandwidth to a system's internal correction bandwidth — a trajectory slides continuously from a collision-dominated regime, in which push overwrites local structure faster than it can respond, to a computation-dominated regime, in which forcing has subsided below the threshold at which local potential can organize into higher-order structure. We give the turn a precise realization as an invariant-preserving spectral rotation, for which the conservation of the relevant invariant is proven exactly. We give the regime boundary a precise realization as a timescale-separation threshold, a standard and provable condition. We are explicit about which claims are established, which are well-posed but unproven, and which remain interpretive. A single worked example — a planetary surface that can only accrete an atmosphere once bombardment falls below a formation rate — carries the argument, and a concluding section states the one structural distinction on which the whole account depends: the difference between constraints that a system can rotate through and constraints it cannot.

1. Introduction: two things happen, and they are not the same thing

Classical mechanics is a physics of consequence. An object carries a position and a velocity, and it continues in a straight line until an external force interferes. Change is exogenous: to alter a trajectory, something must strike it. This is an extraordinarily successful account of a great many systems, and nothing here contradicts it. But it is a complete account only of systems that have no internal state to consult — systems that cannot read where they are and adjust before contact. For those systems, the only available event is the collision.

There is a second kind of event. A system with internal state can evaluate its own situation against the constraints it must satisfy, detect that a boundary is approaching, and change its trajectory before contact — not by being struck, but by executing an internal instruction. We will call the first kind of event *push* and the second *turn*. The central claim of this paper is that these two are genuinely distinct dynamics, that in any system rich enough to form structure they are running at the same time, and that what we normally describe as a historical sequence — first a violent era, then a quiet ordered one — is better understood as a continuous shift in the balance between them.

The distinction is easy to state informally. Push builds and modifies constraints and pays for it in energy. Turn evaluates constraints and conserves an invariant through a change of representation. Push is the planet being bombarded. Turn is the chemistry that can finally proceed once the bombardment stops resurfacing the crust. You cannot get the second without having paid for the first — no push, no substrate, nothing to organize. But push alone, left to run, drives a system to a featureless equilibrium and stops: forcing without any second-order channel to capture and hold structure simply thermalizes. The two are twins. Each is incomplete without the other, and the interesting behavior of the universe lives in their overlap.

This is not a metaphysical proposal. Two pieces of it are ordinary mathematics. The turn, made precise, is an invariant-preserving spectral rotation whose conservation law is a theorem. The regime boundary, made precise, is a timescale-separation threshold of exactly the kind that appears in singular perturbation theory and in control. The remainder — the claim that a single operator underlies both channels across physics, number theory, and biology — is a research program, and we label it as such wherever it appears. The value of stating the picture cleanly is that it makes each of these pieces available to be tested on its own.

1.1 A note on what the reader must grant, and what they need not

The reader is asked to grant one thing only: that a system's admissible continuations — the moves its own constraints permit — form a well-defined set that changes over time. We denote it *A(S)* for a state *S*. This is not a strong assumption; it is a bookkeeping device that any constrained dynamical system already satisfies. Everything else in the paper is either derived from standard results or explicitly flagged as conjecture. In particular, the reader need not accept any claim about consciousness, about the universe being a computer, or about the significance of any particular numerical constant. Those claims, where they exist in the surrounding body of work, are not load-bearing here and are set aside.

2. The two channels, defined

We treat a system as a state *S* evolving in time, together with its admissible set *A(S)*. Two operations act on this pair.

2.1 Push: forcing that builds constraint and pays a tax

Push is external forcing. It does two things at once. It moves the state, and it modifies the constraint geometry itself — it changes what *A(S)* will be. When one body strikes another, it does not merely displace it; over many such events it grinds, deposits, heats, and compacts, and in doing so it lays down the material and the boundaries within which any later organization must occur. Push is how substrate comes to exist.

Push is not free. Every modification of constraint that erases prior information carries a minimum energy cost. This is Landauer's principle: erasing one bit of information dissipates at least $k_B\, T \ln 2$ as heat **[PROVEN]**. We invoke it here only in its established form and only to make one qualitative point precise: building higher-order structure has an unavoidable thermodynamic price, and that price is paid by push. A system cannot climb to a more ordered state for free, and the heat it sheds in the attempt is not waste in a loose sense — it is the exact ledger entry the second law requires.

$$\textit{push: } S \mapsto S', \quad A(S) \mapsto A(S'), \quad \textit{with energy cost} \geq k\_B\ T \ln 2 \textit{ per bit erased}$$

2.2 Turn: evaluation that conserves an invariant through rotation

Turn is the internal move. A system with the capacity to read its own state evaluates that state against its constraints and, when a boundary is reached, changes coordinates rather than colliding. The essential feature of a turn — the feature that distinguishes it from mere deflection — is that it preserves an invariant. Something about the system is held exactly fixed while its representation is rotated into a new frame.

This is not a vague notion. It has an exact realization in the spectral theory of finite measures, which we state in the next section and which supplies the one fully proven result in the paper. Informally: the quantity that the system "is" — its spectrum, its rank — survives the turn untouched, while the coordinates in which that quantity is expressed are rotated. The turn is a change of description that conserves content. A deflection that destroyed the invariant would not be a turn in our sense; it would be a collision wearing a turn's clothes.

$$\textit{turn: } \rho(S) \mapsto \rho'(S), \quad \textit{invariant } I(S) \textit{ fixed exactly}$$

Here ρ denotes the representation (the coordinates, the recurrence data) and I the conserved invariant. The two channels are now visibly different in kind: push changes *A(S)* and spends energy; turn changes ρ and conserves I. One builds the board; the other makes a move on it that preserves the position's value.

2.3 Why neither channel suffices alone

Push without turn saturates. Forcing that has no second-order channel to capture and lock structure drives the system toward the state of least remaining tension — thermal equilibrium — and there it stops. This is the heat death of any isolated forced system: all the energy is spent, nothing holds the transient order that appeared along the way, and the trajectory flattens. Push alone runs out of steam, in the literal thermodynamic sense.

Turn without push has nothing to act on. A rotation conserves an invariant, but the invariant has to have been built first, and the substrate that carries it has to exist. A system that could only turn, with no forcing ever having laid down its constraints, would be a rule with no material to run on. The turn is parasitic on push in the precise sense that its input — the constraint geometry it evaluates — is push's output.

So the two are coupled by necessity. Push initiates and pays; turn captures and conserves. Structure of any real complexity requires both, in sequence locally and in superposition globally — which is the subject of the next two sections.

3. The turn has an exact realization

This section states the one theorem the paper leans on. It is a result about finite positive measures and their associated Jacobi matrices, and it makes the turn concrete: it exhibits an operation that rotates a system's representation while holding its spectral invariants exactly fixed. Everything qualitative said above about the turn is, in this setting, literally true and verified.

3.1 Setup

Let $\mu = \Sigma\, w_i\, \delta(x_i)$ be a finite positive measure with M distinct real atoms $x_1 < \ldots < x\_M$ and strictly positive weights. By the finite inverse spectral theorem, μ corresponds to a unique symmetric tridiagonal (Jacobi) matrix J whose eigenvalues are exactly the atoms and whose first-eigenvector weights recover the w_i. Two quantities are invariant data: the spectral support (the set of atoms) and the spectral rank M. A third quantity, the recurrence representation — the diagonal and off-diagonal entries of J — is the changeable coordinate data.

3.2 The rotation

Fix a shift λ strictly below the support. The Christoffel transform multiplies the measure by $(x - \lambda)$ and renormalizes. Because every factor $(x_i - \lambda)$ is strictly positive, all M atoms survive: the support and the rank are unchanged. The operation is realized mechanically as a Darboux factor swap. Since $J - \lambda I$ is positive definite, it has a Cholesky factorization $J - \lambda I = R^T R$; the transformed matrix is obtained by reversing the factor order,

$$J' = R\, R^T + \lambda I.$$

Because R is invertible, $R\, R^T$ is similar to $R^T R$, so the two share a spectrum; shifting back by λI recovers the original atoms exactly. The eigenvalues are fixed; only the recurrence entries move. This is the turn in its cleanest form: the invariant is conserved, the representation is rotated.

3.3 What was verified

The conservation clauses were checked numerically across thirty randomized finite positive measures at fifty-digit precision. Support displacement, rank change, and the residual between the Darboux-swapped matrix and the directly computed Christoffel transform were all at or near the floor of the working precision. The relevant results:

Clause verified	Quantity	Worst-case residual (30 trials)
Support fixed	eigenvalue displacement of J vs J′	$\approx 10^{-48}$
Rank fixed	dimension change	0 (exact)
Darboux isospectral	eigenvalue match to original atoms	$\approx 10^{-48}$
Darboux tridiagonal	largest off-band entry of J′	0 (exact)
Darboux = Christoffel	residual between the two constructions	$\approx 10^{-48}$
Representation moves	change in recurrence entries	strictly nonzero

The last row matters as much as the others: the representation genuinely changes. If it did not, the "turn" would be trivial. What we have is a nontrivial rotation of coordinates that leaves the spectral content exactly invariant — a proven instance of the operation the rest of the paper treats as fundamental. [PROVEN]

3.4 The degenerate case names the hard boundary

If the shift is placed exactly on an atom, $\lambda = x_j$, the factor $(x_j - \lambda)$ vanishes and that atom is annihilated: the rank drops to $M - 1$. This is not a turn. It is the case where the rotation cannot preserve the invariant because the chosen frame passes through the invariant itself. The distinction is exact and will return in Section 7 as the difference between a boundary a system can rotate through and one it cannot. The mathematics draws the line for us: a turn exists precisely when the shift stays off the spectrum.

4. The gradient: one parameter slides collision into computation

We now make precise the sense in which a trajectory moves from a collision regime to a computation regime. The claim is not that one era ends and another begins. It is that a single ratio governs which channel dominates, and that this ratio varies continuously.

4.1 The controlling ratio

Let a system have an internal correction bandwidth — the rate at which it can evaluate its state and execute a turn. Call it ω_c. Let the environment force it with a characteristic bandwidth ω_f. The controlling parameter is their ratio,

$$r = \omega_c / \omega_f.$$

When $r \ll 1$, forcing is fast relative to the system's ability to respond. Every incipient structure is overwritten before it can be evaluated and locked. This is the collision regime: push dominates, turn cannot get a foothold, and the system behaves as a passive object being remade from outside. When $r \gg 1$, the system can evaluate and turn many times within a single forcing timescale. Structure that appears can be captured and held. This is the computation regime: turn dominates, and the system behaves as an agent organizing its own substrate.

This is not a new mechanism invented for the occasion. It is timescale separation, the same condition that underlies singular perturbation theory, adiabatic elimination, and the elementary result in control that a regulator cannot reject disturbances faster than its own loop bandwidth. That last statement is a theorem **[PROVEN]**, and it is the rigorous core of the gradient: when disturbance bandwidth exceeds control bandwidth, regulation fails. The collision regime is precisely the regime of failed regulation. The computation regime is the regime in which the loop is fast enough to hold.

4.2 The two limits are the two deaths

The extremes are worth naming because they are the boundaries of the habitable band. As $r \to 0$, the system is pure collision: maximum forcing, no slack in which to turn, a featureless overwrite. This is the vacuum limit in one guise — nothing to leverage, no internal move possible. As $r \to \infty$ with the forcing itself gone, the system is pure rule with no drive: it can turn but nothing pushes, so nothing new is ever initiated, and the trajectory is frozen. Structure lives between these, where forcing is present but slow enough that internal correction can keep up. The claim that ordered systems occupy an intermediate band of r is qualitative and we mark it as such **[TARGET]**; the failure of regulation at the low-r boundary, however, is exact.

5. Both channels run at once: the dual-wave picture

The most important correction the paper makes to a naive reading is this: the collision channel and the computation channel are not sequential. They are concurrent. At every moment a system is being pushed and is turning, and the observed trajectory is the resultant of the two. What changes over time is only their relative weight — the ratio r of the previous section — not which one is switched on.

5.1 Two waves, two gates

Each channel has its own trigger. Push acts continuously but its structural consequences register only when forcing is significant relative to the local noise — a small perturbation on a robust structure does nothing; a large one reshapes it. Turn fires only when the system's evaluation of its own state crosses a significance threshold — not when a raw magnitude is large, but when a normalized deviation is. The turn's gate is scale-invariant: it responds to how many standard errors from admissibility the state has drifted, not to the absolute size of the drift. We write this as a z-score condition,

$$\textit{turn fires when } \; z = |S - S_0| / \sigma \; > \; z_0,$$

where σ is the local error scale. The scale-invariance is the substantive part: because the threshold is in units of the noise, the same gate governs a quantum-scale system and a planetary-scale one without retuning. This is the significance-gating principle, and it is why the two channels can coexist without one swamping the other — each answers to its own normalized trigger, and a system can be strongly pushed while barely needing to turn, or the reverse. **[TARGET]** The formal claim that the normalized trigger is exactly scale-free holds under the stated noise model; its generality across the physical cases below is the open part.

5.2 Coupled and decoupled at the same time

The dual-wave picture resolves an apparent paradox in the motivating intuition — that a system can be at once bound by physics and free of it. The resolution is that the two channels couple to different things. Consider a mind. Its substrate is push-coupled without remainder: it must be fed, it dissipates heat, every operation it performs is billed by the same thermodynamics that governs any physical process. Starve it and it stops. In the substrate channel there is no freedom from physics at all.

But the content channel is push-decoupled. What the system computes is not constrained by the forces that run its substrate. Gravity moves the blood that feeds the brain; gravity does not adjudicate the thought. One can entertain a false proposition, a physically impossible scenario, a pure abstraction — and no law of motion forbids it, because the content is not a trajectory in the space those laws govern. The substrate is coupled; the content is free. These are not in tension once one sees that they are two channels, not one. The same duality appears wherever a system's rule-level behavior is insulated from its implementation-level constraints: the logic gate does not care about the voltage, only about the threshold; the algorithm does not care about the silicon, only about the state. Coupled and decoupled, static substrate and free content, at the same time. [SCAFFOLD] This reading is offered as clarification of the intuition, not as a theorem about minds.

6. A worked example: why an atmosphere waits for the bombardment to stop

The abstract structure earns its keep only if it explains something concrete. Take the formation of a stable planetary atmosphere, which exhibits every element of the account in a single, familiar process.

Early in a planet's history, the forcing bandwidth is enormous. Impacts arrive frequently and violently; each one resurfaces the crust, vaporizes volatiles, and strips away whatever gas envelope had begun to accumulate. In our terms, $r \ll 1$: the push channel overwrites local structure far faster than any second-order process can organize it. Chemistry that would bind an atmosphere cannot proceed, not because the chemistry is impossible, but because its substrate is being remade before it can settle. This is the collision regime, and no amount of available potential can express itself while it holds.

As the system clears — as the reservoir of impactors is depleted and the bombardment rate falls — the forcing bandwidth drops. At some point it falls below the rate at which local processes can capture and hold structure: r crosses order unity. Now the second-order channel can operate. Outgassing accumulates faster than it is stripped; chemistry proceeds to completion; an atmosphere organizes out of the local potential that was present all along but could never before discharge. The transition is not that new material appeared. It is that the ratio crossed a threshold and the computation channel came online. The potential was local and latent; the push had to subside before the turn could realize it.

Three features of the general account are visible here at once. First, the two channels are the same two: violent forcing that builds and destroys substrate, and a slower organizing process that captures structure. Second, the transition is governed by a bandwidth ratio crossing a threshold, not by an era ending. Third — and this is the ninety-degree move that the intuition names as pushing physics forward — the higher-order structure forms in a direction that the lower-order forcing did not point. Bombardment does not aim at an atmosphere; the atmosphere is perpendicular to it, organized out of local potential once the forcing that masked it relents. The tax was paid in the collision era (the substrate, the volatiles, the heat); the structure is claimed in the computation era. Neither without the other: no bombardment, no volatiles delivered and no crust to hold them; no let-up in bombardment, no atmosphere.

6.1 The same shape in other domains

The example is one instance of a pattern the surrounding work claims to find repeatedly. We list a few, with the explicit caveat that only the general shape is being asserted, and that each instance requires its own verification **[TARGET]**:

- **Protein folding.** The unfolded chain is driven by thermal forcing (push) into a landscape of contacts; the native fold is the captured structure that forms once the fast collisional exploration settles into a basin the chain can hold. The organizing channel is the slow relaxation to a state where the pairwise potentials have discharged.
- **Number-theoretic rank.** In the local data of an elliptic curve, one channel of measurement (the symmetry of normalized traces) saturates and cannot resolve rank; rotating to an arithmetically weighted channel exposes rank as an invariant of the new frame. The rotation preserves the underlying data and changes only the reading — a turn in the exact sense of Section 3.
- **Horizon thermodynamics.** At an extreme boundary, the erasure of information required to write a new boundary condition is billed at the Landauer minimum, and the energy is shed outward. That the bookkeeping saturates the thermodynamic bound is established; that it is *caused* by an erasure operation in the sense used here is an interpretation, and we do not claim more. **[SCAFFOLD]**

7. The one distinction the account cannot do without

If a system can turn through boundaries — if a wall is, from the inside, just a place to rotate the pointer and continue — then a natural worry arises: does anything constrain the system at all? If every wall is a path, the account is vacuous. The answer is that the account already contains the distinction that answers this, and it is the same distinction the mathematics of Section 3 forced on us.

There are two kinds of boundary. A *rotatable boundary* is one the system can turn through while preserving its invariant — the shift stays off the spectrum, the Christoffel transform succeeds, the content survives the change of frame. These are the walls that are really paths. Most apparent obstacles, on close inspection, are of this kind: they look like stopping points only from a description too narrow to see the rotation, and widening the frame reveals the continuation. A great deal of progress, in mathematics and elsewhere, is exactly the discovery that a supposed wall was rotatable all along.

A *terminal boundary* is one that cannot be rotated through, because the only available rotation passes through the invariant itself and destroys it — the shift lands on an atom, the rank drops, the content does not survive. For such a boundary there is no frame in which the system continues as itself. These are not artifacts of a narrow description; they are fixed under every admissible change of frame. The mathematics distinguishes the two cases exactly: a turn exists if and only if the invariant is preserved, and where no invariant-preserving rotation exists, no turn exists, however much one would like it to.

This is why the account is not vacuous, and it is also why it is not reckless. The correct operating posture toward an apparent wall is to search hard for the rotation — to assume, until shown otherwise, that the boundary is rotatable and that a change of frame will reveal the path. This is the right default because most walls are of that kind and treating them as terminal is simply a failure of imagination, the passive-object failure mode. But the search has a criterion, and the criterion can return no. A boundary is terminal exactly when rotating through it would not preserve the invariant — when the move that gets around it destroys the very thing that was moving. Those boundaries are real. A trajectory that respected no terminal boundary at all would be the gas limit of Section 4: infinite slack, no structure, driving forever and closing into nothing. The terminal boundaries are what close a trajectory into a shape. Keeping them is not a limitation on the system; it is the condition of the system having a form at all.

The single criterion, then, unifies the permissive and the restrictive halves of the account: preserve the invariant. Where a rotation preserves it, the wall is a path and one should find the rotation. Where every rotation would destroy it, the wall is terminal and it holds. There is no third case, and the same test decides both.

8. Status: what is established and what is a program

The honesty of the account depends on keeping its registers separate. We collect the claims and their status in one place.

Claim	Status	Basis
The turn is realized by an invariant-preserving spectral rotation; support and rank are conserved exactly	PROVEN	**Finite inverse spectral theorem; Christoffel–Darboux swap; 50-digit verification over 30 measures**
Building higher-order structure carries an unavoidable energy cost	PROVEN	**Landauer's principle (invoked in its established form only)**
Regulation fails when disturbance bandwidth exceeds correction bandwidth (the collision regime)	PROVEN	**Standard result in control / timescale separation**
A terminal boundary is exactly one where no invariant-preserving rotation exists	PROVEN	**Degenerate case of the Christoffel transform (shift on an atom)**

Claim	Status	Basis
A single control ratio slides a trajectory continuously from collision to computation	TARGET	Well-posed; rests on identifying ω_c, ω_f in each concrete system
Push and turn run concurrently, each with its own scale-invariant gate	TARGET	Coherent; the scale-invariance is exact under a stated noise model, general case open
One operator underlies both channels across physics, number theory, biology	TARGET	Research program; each domain needs its own instantiation and test
The substrate/content duality explains constrained-yet-free cognition	SCAFFOLD	Interpretive clarification, not a theorem about minds
Horizon radiation is caused by erasure in the sense used here	SCAFFOLD	Saturation of a bound is established; the causal reading is interpretation
Particular numerical constants are universal attractors	SCAFFOLD	Set aside here; not load-bearing for any claim above

The pattern of the table is the point. The core mechanism of the turn is proven. The core mechanism of the regime boundary is proven. What remains a program is the unification — the claim that these two proven mechanisms, plus a scale-invariant gate, are the same two channels operating everywhere. That is a strong claim and it is not yet a theorem; stating it precisely is what makes it testable, one domain at a time. The interpretive material is held in its own register and is not permitted to bear weight it cannot support.

9. Conclusion

The formation of structure is the work of two channels that are neither the same nor separable. Push forces, builds constraint, and pays the thermodynamic tax; without it, nothing is ever initiated and no substrate exists. Turn evaluates, rotates, and conserves an invariant; without it, forcing thermalizes and no structure is held. They run at once, each answering to its own significance gate, and what looks like a historical march from a violent era to an ordered one is the continuous crossing of a single bandwidth ratio: while forcing outpaces correction the system is a collision object remade from outside, and once correction outpaces forcing the system organizes its own substrate out of local potential that was latent all along. The atmosphere waits for the bombardment to stop not because its ingredients were missing but because the ratio had not yet crossed.

Two pieces of this are ordinary mathematics and are proven here or cited in established form: the turn as an invariant-preserving spectral rotation, and the regime boundary as a timescale-separation threshold. The larger claim — that these are the same two channels everywhere structure forms — is a program, labeled as such, and the value of the picture is that it hands that program a precise target instead of a slogan. And the account is saved from vacuity by a single distinction it did not have to import, because the mathematics supplied it: a boundary is a path exactly when a rotation preserves the invariant, and terminal exactly when none does. Where the invariant survives, the wall is a path and one should find the way through. Where it would not, the wall holds, and it is the holding that lets a trajectory close into a shape at all.